Introduction to Time Series

- A (univariate) time series is a set of observations of the same variable made sequentially over time.
  - Discrete-time vs. continuous-time time series. We will focus on the former and consider time indices $\mathcal{T} = \{1, 2, \ldots\}$, where $\mathcal{T}$ is the index set. Most often, the time indices represent regular time intervals.
  - Notation: y_t denotes the value of some variable at time t. A time series can be denoted by $\{y_1, y_2, \ldots, y_T\}$, $\{y_t\}_{t=1}^{T}$, or $\{y_t\}$, $t = 1, 2, \ldots, T$, where T is the sample size.
  - y_t is a *random variable!* The observed sample consisting of T particular numbers is just one set of possible outcomes that the underlying stochastic process generated.
  - A *stochastic process* is a collection of random variables indexed by time, $\{y_t : t \in \mathcal{T}\}$. The observed values of a stochastic process are called a *realization* of the stochastic process.
  - Two simple examples
    - Bernoulli process: repeated coin flipping
    - Gaussian white noise: $y_t = \varepsilon_t$, for $t = 1, 2, \ldots, T$, where $\varepsilon_t \sim IIDN\left(0, \sigma^2\right)$.
- Examples of economic time series:
  FRED (Federal Reserve Economic Data: `https://fred.stlouisfed.org/`)
  - Real GDP
  - Real GDP growth
  - Interest rate on 3-month Treasury bills
  - Yield on 10-year Treasury bonds
  - Unemployment rate
  - Housing starts
- Important features of time series data
  - Serial correlation/non-independence of observations: y_t depends in some way on $y_{t-1}, y_{t-2}, \ldots$
  - Ordering matters! Changing the order could change the meaning of the data.
- Time series analysis is concerned with modeling the dependency of a variable on its own past, and on the current and past values of other variables.

- Objectives of time series analysis
  - To describe important features/patterns
  - To estimate dynamic causal effects
  - To forecast future values
  - To model risks in financial markets
- It is a good habit to begin by plotting a time series and consider the following questions
  - **Trend**? A persistent, long-term movement or tendency. A trend need not be a straight line!
  - **Seasonality**? Periodic fluctuations due to calendar/seasonal factors—high/low values tend to occur in some particular months/quarters/days of the week. Seasonality is always of fixed and known period.

    Causes of seasonality:
    - Weather-related factors (e.g. temperature, hours of sunshine)
    - Calendar effects (e.g. timing of religious festivals/holidays, such as Christmas and Easter)
    - Timing decisions (e.g. school and industry vacations, tax years, accounting periods, dates for dividend or bonus payments)
  - **Long-run cycle**? Cyclic patterns where the data exhibits rises and falls that are NOT of fixed period. Duration of fluctuations is usually $\geq$2 years.
  - **Constant variance**?
  - **Abrupt change** in the level or variance?
  - **Outlier**? Data point that is far away from other data